

Southern Ocean

The icy Southern Ocean surrounding Antarctica is the windiest, most dangerous ocean on the planet. It is scattered with towering icebergs that have broken off Antarctica's vast ice sheets and glaciers, and in winter its surface is a sea of tumbled pack ice. Cold water flowing from beneath the ice drives powerful deepwater currents that travel all the way around the world.

FACT FILE

Area:	13,514,000 sq miles (35,000,000 sq km)
Average depth:	11,000 ft (3,350 m)
Deepest point:	23,740 ft (7,235 m)

Weddell and Ross seas



▲ FLOATING ICE

The floating tip of a giant glacier towers above the fractured pack ice in the Ross Sea.

The enormous bays of Weddell and Ross seas divide the southern continent into West and East Antarctica. The ocean water they contain lies close to the South Pole, and much of it is covered by gigantic floating ice shelves and glaciers. The rest freezes over in winter, to become seas of drifting pack ice.



Howling winds

Powerful winds blow from west to east over the Southern Ocean throughout the year, because there is no land to slow them down. They get stronger the further south they are, reaching storm force near Antarctica. These winds were a big help to ocean trade in the days of sailing ships, driving them around the globe.



▲ OCEAN RACERS

Modern sailboats use powerful Southern Ocean winds to race around the world.

Rich waters

The ocean's northern limit is known as the Antarctic Convergence, where very cold water sinks beneath the warmer Pacific, Atlantic, and Indian oceans. This encourages the growth of plankton that feed swarms of shrimplike krill, which in turn are eaten by other animals.

▲ SUMMER VISITORS

These Arctic terns have flown halfway around the world to feed in the rich Antarctic seas.

► FROZEN OCEAN

In winter the Southern Ocean around Antarctica freezes over, creating a vast expanse of pack ice (pale blue). The white dotted line marks the boundary of the ocean at the Antarctic Convergence.

